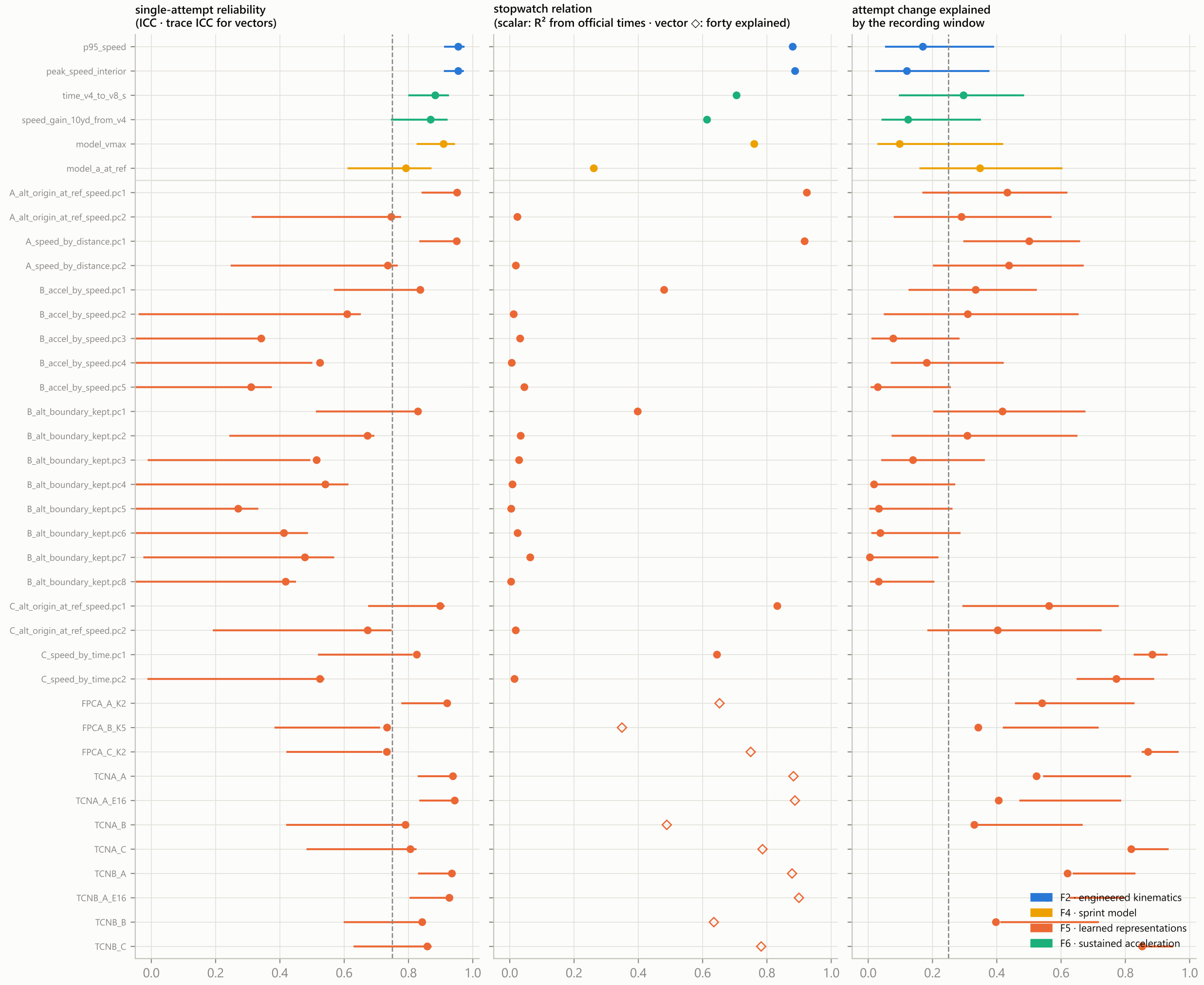
Functional PCA and encoders in the feature-characterisation context



Unit: athlete. Above the line: the two most reliable hand-defined candidates of the kinematic, sustained-acceleration and sprint-model families, for scale. Below: functional PCA components, and each representation as a whole vector (FPCA K scores, TCN-A, TCN-B). Estimators differ by design and are named in atlas_characteristics: a scalar has an ICC, a vector a rotation-invariant trace reliability; the middle panel shows how much of the scalar the official times explain, and for a vector how much of the official forty the vector explains (held out, beyond position). Bars = 95 %.